

Mathematics Extended Essay

Elliptic Curve Cryptography

How can elliptic curves be used to achieve online security using the Elliptic-Curve Diffie-Hellman protocol?

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Table of Contents

1. Introduction	1
2. Weierstrass Equations	2
3. Point Addition	3
4. Finite Fields	8
5. Elliptic Curves as Groups	13
6. Group Properties	15
7. Domain Parameters of ECDH	17
8. Diffie-Hellman Key Exchange	18
9. Elliptic Curve Discrete Logarithm Problem	19
10. Selecting Domain Parameters	20
11. Conclusion	23

1. Introduction

Elliptic curves have many applications across cryptography, namely they are used in the Elliptic-Curve-Diffie-Hellman (ECDH) protocol, which is in turn used in the Signal protocol implemented in Signal, WhatsApp, FaceBook Messenger, and Skype to send and receive encrypted messages (Signal, 2023). These encrypted messages can only be read by the sender and receiver, thus they are secure to third-party intercepts when the messages are sent online. This is important as it allows people to send private messages publicly, specifically over the internet.

ECDH is not the only protocol that can be used for establishing this level of security; the Rivest-Shamir-Adleman (RSA) algorithm is a great example, as it is the most well-known modern cryptography system (Rivest, Shamir, Adleman, 1978). So why choose to cover ECDH? Other than it being used in applications used by millions daily (RSA is too), it also allows for exploration of more abstract mathematics, more interesting for investigation. Elliptic curves themselves also have further applications to new quantum-proof encryption systems, thus looking at ECDH is more relevant to the future of cryptography.

In this essay I will begin by establishing the mathematical concepts required for the ECDH protocol (sections 2-6). I will continue on to describing the protocol itself (sections 7-9) and finish by evaluating some weaknesses of the protocol (section 10), namely the quantum computing revolution.

2. Weierstrass Equations

The first mathematical concept important for understanding ECDH is understanding elliptic curves themselves and a number of their properties. It is not necessary to formally define elliptic curves, as we are only interested in elliptic curves over fields $\mathbb{F}$ of characteristic $\neq 2, 3$, which simplify to the Weierstrass equation:

$$y^2 = x^3 + Ax + B \quad (1)$$

For any A, B in $\mathbb{F}$ (Koblitz, 1987). It is not important to have an understanding of fields and field characteristics as they will be described later in section 4.

We are also only interested in non-singular elliptic curves, i.e. without a multiple root (Dummit, 2016), so the discriminant $\Delta \neq 0$. Δ can be derived from the cubic on the right-hand side of equation (1) and is calculated as follows:

$$\Delta = -16(4A^3 + 27B^2)$$

To visualize equation (1) a few examples are shown below in Figure 1.

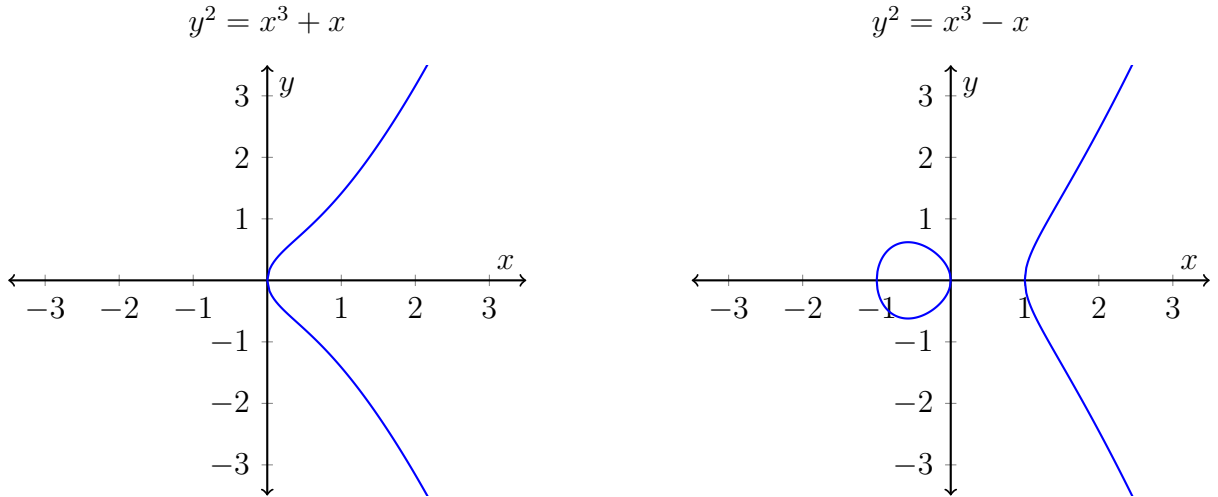


Figure 1: Example Elliptic Curves Over $\mathbb{Q}$

Hence we can define an elliptic curve over a field $\mathbb{F}$ with characteristic $\neq 2, 3$ as the set of points $(x, y) \in \mathbb{F}^2$ that satisfy equation (1) for any $A, B \in \mathbb{F}$, $\Delta \neq 0$.

3. Point Addition

Having established an equation for elliptic curves of characteristic $\neq 2, 3$, now the operation of point addition will be defined.

For this a point at infinity, denoted $\mathcal{O}$, must also be defined according to Housni (2018). It is convenient to imagine this point at infinity as the 'top' or 'bottom' of the y -axis. So a vertical line passing through an elliptic curve E passes through either: a single point, if the y coordinate is 0, or two points otherwise, and the point at infinity $\mathcal{O}$. If a line is not vertical, it will pass through either two or three points on E and the point at infinity $\mathcal{O}$. This can be written as:

$$P + Q + R = \mathcal{O} \tag{2}$$

The first approach to adding points on an elliptic curve is geometric, demonstrated by Housni (2018). Let the points $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be points on the elliptic curve E defined by $y^2 = x^3 + Ax + B$. To add $P + Q$ we simply draw the line through both points and find the third intersect with E and can denote it as $-R$. Then we reflect that point on the x -axis to get the point R . This reflection can always be done as E is symmetrical about the x -axis ($y^2 = ..$).

When attempting this method with various points on E there are still a few concerns. Firstly, how do we know that there will be an intersect between our drawn line and E ?

If P and Q are points with distinct x coordinates, the line between them will not be vertical, thus substituting the equation of our line into E gives a cubic with three distinct roots, and thus must intersect three times.

Else if P and Q are distinct, but have the same x coordinate, our line will be vertical. In this case it does not intersect E a third time, instead we let it intersect E at the point at infinity $\mathcal{O}$. In this case we can also let $Q = -P$, which is useful in the next section.

The final concern is what to do if $P = Q$. In this case, instead of drawing the line between two points, we draw the tangent line, and continue from there as before. If both points have a y coordinate of 0, this tangent line is again vertical, and hence will intersect at $\mathcal{O}$.

For example, using the elliptic curve of the form $y^2 = x^3 - x$ over $\mathbb{Q}$, we can add two distinct points P and Q geometrically as shown below.

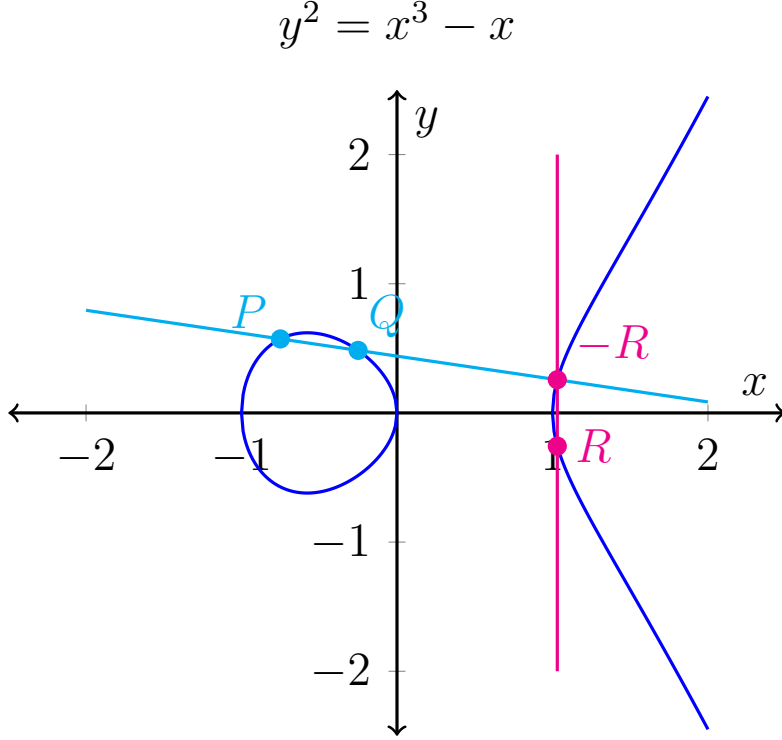


Figure 2: Example of Point Addition on $y^2 = x^3 - x$ Over $\mathbb{Q}$

If any of the special cases above occurred, the method would need to be altered accordingly, however, this should not need more examples.

It is also important to be able to perform point addition algebraically, as this is how a computer program would approach the operation. This follows neatly from the geometric approach.

In the first case, where $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ are points with distinct y coordinates, we calculate the line passing through them. If $P = Q$ and P, Q do not have y coordinates of 0 we calculate the tangent line. Note that $P + P$ can be written $2 \cdot P$. These cases only differ in the calculation of the gradient of the line. For distinct points it is calculated:

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad (3)$$

And for a tangent line it is calculated using implicit differentiation:

$$\begin{aligned}
y^2 &= x^3 + Ax + B \\
\Rightarrow 2y \frac{dy}{dx} &= 3x^2 + A \\
\frac{dy}{dx} &= \frac{3x^2 + A}{2y} \\
m = \frac{dy}{dx} &= \frac{3x_1^2 + A}{2y_1} = \frac{3x_2^2 + A}{2y_2}
\end{aligned} \tag{4}$$

Next, the y -intercept c can be calculated using the gradient m and coordinates of either of the points:

$$\begin{aligned}
y_1 &= mx_1 + c \\
c &= y_1 - mx_1 = y_2 - mx_2
\end{aligned} \tag{5}$$

Giving an equation of the line:

$$y = mx + c \tag{6}$$

By substituting equation (6) into equation (1), the equation of E we find the intercept with the curve E :

$$\begin{aligned}
(mx + c)^2 &= x^3 + Ax + B \\
m^2x^2 + 2cmx + c^2 &= x^3 + Ax + B \\
x^3 + (-m^2)x^2 + (A - 2cm)x + (B - c^2) &= 0
\end{aligned} \tag{7}$$

The cubic equation (7) has three roots. Since we know two of the roots are x_1 and x_2 we can find x_3 (the x coordinate of $-R$ and R) using the rule for the sum or product of the roots of a polynomial (known as Vieta's equations). Using the rule for the sum is simpler:

$$\begin{aligned}
x_1 + x_2 + x_3 &= \frac{-(-m^2)}{1} \\
x_3 &= m^2 - x_1 - x_2
\end{aligned} \tag{8}$$

Next the y coordinate of $-R$ is found by substituting x_3 into equation (6):

$$y_3 = mx_3 + c \tag{9}$$

Finally, to reflect $-R$ on the x -axis we change the sign of y_3 , giving $R = (x_3, -y_3)$.

Now, if instead we are adding two points with the same x coordinate, our third point will always be $\mathcal{O}$, so no algebraic method needs to be established. The syntax of two points with the same x coordinates, however, is important later. If two points P and Q on E have the same x coordinate, but different y coordinates we express $Q = -P$, and the final equation is written:

$$P + (-P) = \mathcal{O} \quad (10)$$

If both points are equal and have y coordinates of 0, $P = -P$, so according to (10) we can write:

$$P + P = 2 \cdot P = \mathcal{O}$$

Now to return to our geometric example in Figure 2. The points $P = (-0.75, 0.572821961869)$ and $Q = (-0.25, 0.484122918276)$ respectively (x -coordinates chosen by student). Using the coordinates of both points we can calculate the gradient for the line passing through them using equation (3):

$$\begin{aligned} m &= \frac{0.484122918276 - 0.572821961869}{(-0.25) - (-0.75)} \\ &\approx 0.177398087186 \end{aligned}$$

Next, the constant term c of the line is calculated using equation (5):

$$\begin{aligned} c &= (0.572821961869) - (0.177398087186)(-0.75) \\ &\approx 0.439773396479 \end{aligned}$$

Hence the line passing through both P and Q is:

$$y = 0.177398087186x + 0.439773396479 \quad (11)$$

Our next step is to find the x coordinate of the intersect of equation (11) with our curve using equation (8):

$$\begin{aligned} x_3 &= (0.177398087186)^2 - (-0.25) - (-0.75) \\ &\approx 1.03147008133 \end{aligned}$$

And substitute x_3 into (11) to get the y coordinate of $-R$:

$$\begin{aligned}y_3 &= 0.177398087186(1.03147008133) + 0.439773396479 \\ &\approx 0.25679257706\end{aligned}$$

Hence, changing the sign of y_3 , we get $R = (1.03147008133, -0.25679257706)$.

Before ending this section, scalar multiplication must also be defined according to Rivain (2011). This is the reason why we can write $P + P = 2 \cdot P$. Using this, we can 'multiply' a point P by an integer number n by adding it to itself multiple times. The efficiency of this can be increased by always adding the highest multiple of 2 possible, for example to calculate $9 \cdot P$, we calculate $P + P = 2 \cdot P$, then $2 \cdot P + 2 \cdot P = 4 \cdot P$, then $4 \cdot P + 4 \cdot P = 8 \cdot P$, and finally $8 \cdot P + P = 9 \cdot P$. This is called the double-and-add algorithm (Rivain, 2011).

4. Finite Fields

In sections 2 and 3 it has been stated that elliptic curves over a field of characteristic $\neq 2, 3$ simplify to the Weierstrass equation (1). To understand ECDH we do not need to understand the formal definition of fields and field characteristic; we must, however, note that the field of rational numbers $\mathbb{Q}$ has a characteristic of 0, so our initial examples and understanding of point addition on elliptic curves are valid.

Next, the reason why fields of characteristic $= 2, 3$ are not relevant is because in ECDH, elliptic curves over prime finite fields $\mathbb{F}_p$ are used, where p is a prime. The characteristic of these fields $= p$ (Weisstein, 2023), and for ECDH much larger primes than 2 and 3 are used (explained in section 9).

A prime finite field can be described as the field of integers modulo p or $\mathbb{Z}/p\mathbb{Z}$ (Housni, 2018). To understand what this means we need to look briefly at linear congruences defined by Blythe, Joseph, Urban, R. Haese, M. Haese (2005). Let $a, b, m \in \mathbb{Z}$. a and b are congruent modulo m if they leave the same remainder when divided by m . This is written as:

$$a \equiv b \pmod{m}$$

a is called a residue modulo m . A complete system of residues modulo m is the set of integers $0, 1, \dots, m-1$, which is to say that all residues can be reduced to a number within the set, which is called the least positive residue.

We can perform the operations of addition and multiplication modulo m identically to regular arithmetic. Division does cause some problems, however, we will not need it. In modular arithmetic it is best practice to always reduce numbers to their least positive residues when possible. To do this we can add or subtract multiples of m . For example:

$$\begin{aligned} 15 + 4 \pmod{7} &= (15 - 2(7)) + 4 \pmod{7} \\ &= 1 + 4 \pmod{7} \\ &= 5 \pmod{7} \end{aligned}$$

Or multiplication:

$$\begin{aligned}
12 \times -9 \pmod{7} &= (12 - (7)) \times (-9 + 2(7)) \pmod{7} \\
12 \times -9 \pmod{7} &= 5 \times 5 \pmod{7} \\
&= 25 \pmod{7} \\
&= 25 - 3(7) \pmod{7} \\
&= 4 \pmod{7}
\end{aligned}$$

An elliptic curve E over a finite field $\mathbb{F}_p$ of characteristic $\neq 2, 3$ behaves almost identically to an elliptic curve over $\mathbb{Q}$. The most obvious difference is that now points on the curve are restricted to integers $< p$. The equation of E is altered to:

$$y^2 \equiv x^3 + Ax + B \pmod{p} \quad (12)$$

where $A, B \in \mathbb{F}_p$ and $A, B \not\equiv 0 \pmod{0}$ (Housni, 2018). Examples of what this can look like are shown in Figure 3 below.

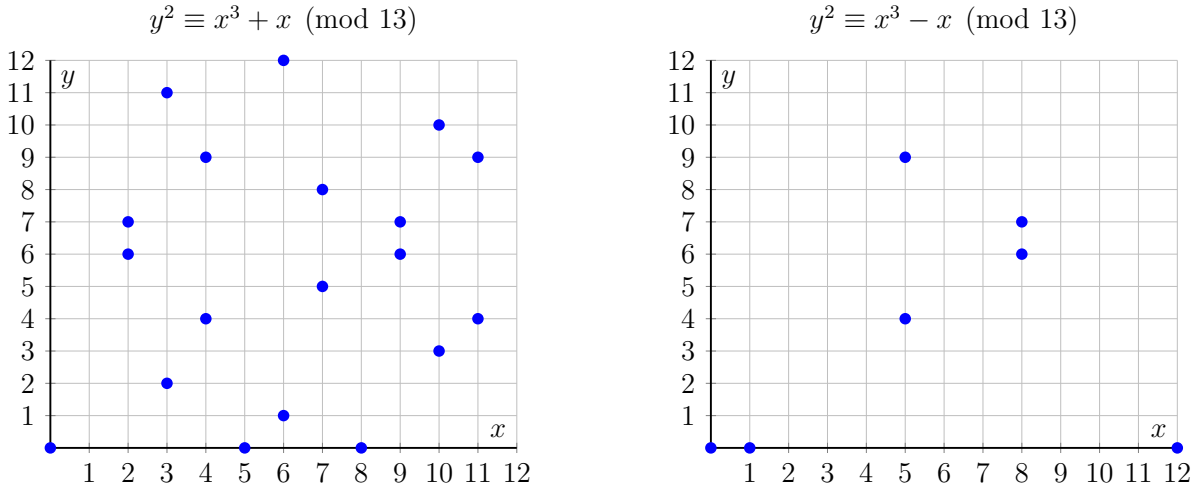


Figure 3: Example Elliptic Curves Over $\mathbb{F}_{13}$

The next detail that must be noted is that now E has symmetry about $y = \frac{p}{2}$ instead of $y = 0$. This does not seem to apply to points where $y = 0$, which is reasonable considering $0 \equiv p \pmod{p}$, so our point is symmetrical of itself, similar to points at $y = 0$ on elliptic curves over $\mathbb{Q}$.

The operation of point addition still works on elliptic curves over finite fields, but now our line between two points will be of the form $y \equiv mx + c \pmod{p}$. This can be geometrically

represented as a line, which when meeting the edges of the finite field will wrap back to the opposite edge (Housni, 2018). The edges need to be changed to p instead of $p - 1$ for this to work, despite points at p being equal to points at 0, but we can imagine this as still excluding p . An example of point addition on elliptic curves over finite fields is shown below in Figure 4.

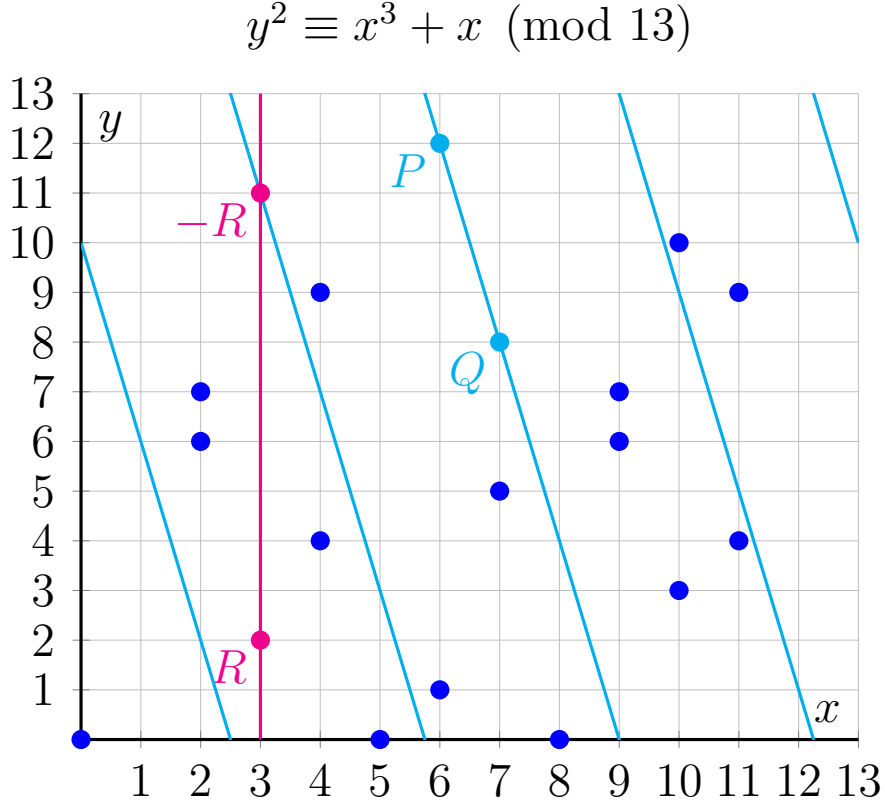


Figure 4: Example of Point Addition on $y^2 \equiv x^3 + x$ Over $\mathbb{F}_{13}$

Finally, we must once again be able to perform point addition algebraically. The steps of doing so are identical as with elliptic curves over $\mathbb{Q}$, however, each calculation is preformed $(\text{mod } p)$. So first calculating the gradient from equation (3) for lines between two points we have:

$$m = \frac{y_2 - y_1}{x_2 - x_1} \pmod{p} \quad (13)$$

And from equation (4) for a tangent line we have:

$$m = \frac{dy}{dx} = \frac{3x_1^2 + A}{2y_1} = \frac{3x_2^2 + A}{2y_2} \pmod{p}$$

Next we calculate the y -intercept using equation (5):

$$c = y_1 - mx_1 = y_2 - mx_2 \pmod{p} \quad (14)$$

Giving our equation of a line as:

$$y \equiv mx + c \pmod{p} \quad (15)$$

Next, we calculate the intercept between this line and E by substituting equation (15) into equation (12) similarly to equation (7), knowing two of three roots again, we get similarly to equation (8):

$$x_3 = m^2 - x_1 - x_2 \pmod{p} \quad (16)$$

Then, we calculate the y coordinate using equation (15) similarly to equation (9) to get:

$$y_3 = mx_3 + c \pmod{p} \quad (17)$$

Finally, instead of reflecting the point over $y = 0$, we reflect it over $y = p/2$, its new symmetrical axis, to get a final point of $(x_3, y_3 - 13)$ if $y_3 > \frac{p}{2}$ or $(x_3, y_3 + 13)$ if $y_3 < \frac{p}{2}$.

To return to our example from Figure 4 above, the points $P = (6, 12)$ and $Q = (7, 8)$ (x -coordinates chosen by student) over $y^2 \equiv x^3 + x \pmod{13}$. First we calculate the gradient using equation (13):

$$\begin{aligned} m &= \frac{8 - 12}{7 - 6} \pmod{13} \\ &= -4 \pmod{13} \\ &= 1(13) - 4 \pmod{13} \\ &= 9 \pmod{13} \end{aligned}$$

Then the y -intercept using equation (14):

$$\begin{aligned} c &= 12 - (9)6 \pmod{13} \\ &= 12 - 54 \pmod{13} \\ &= 12 - (54 - 4(13)) \pmod{13} \\ &= 12 - 2 \pmod{13} \\ &= 10 \pmod{13} \end{aligned}$$

Giving an equation of a line according to equation (15) as:

$$y \equiv 9x + 10 \pmod{13}$$

Calculating the intercept to get x_3 according to equation (16) we get:

$$\begin{aligned} x_3 &= 9^2 - 6 - 7 \pmod{13} \\ &= 81 - 13 \pmod{13} \\ &= (81 - 6(13)) + (-13 + 1(13)) \pmod{13} \\ &= 3 \pmod{13} \end{aligned}$$

Finally calculating y_3 using equation (17) we get:

$$\begin{aligned} y_3 &= (9)(3) + 10 \pmod{13} \\ &= 27 + 10 \pmod{13} \\ &= (27 - 2(13)) + 10 \pmod{13} \\ &= 1 + 10 \pmod{13} \\ &= 11 \pmod{13} \end{aligned}$$

Reflecting y_3 , since $y_3 > \frac{13}{2}$ on $y = \frac{13}{2}$, gives $13 - y_3 = 2$. Giving a final point $R = (3, 2)$, which is the same as shown on Figure 4.

We have established what finite fields are and how elliptic curves over finite fields behave, including how the point addition operator still functions. The next two sections move on to describe what this implies.

5. Elliptic Curves as Groups

The set of points on an elliptic curve E over a finite field $\mathbb{F}_p$ together with the point at infinity $\mathcal{O}$ form an Abelian group over point addition. This group can be written as $(E, +)$, where

$$E = \{(x, y) \in \mathbb{F}^2 \mid y^2 = x^3 + Ax + B\} \cup \{\mathcal{O}\}$$

and $+$ is point addition, described in the previous section.

A group $(G, \cdot)$ is an abstract algebraic structure, where G is a non-empty set and G is closed over $\cdot$, some binary operator (Note: in this and the following section, $\cdot$ does not denote scalar multiplication of point addition $+$). Knowledge of sets is assumed, and a binary operator is simply an operator which takes in two inputs and returns one output (Blythe, Joseph, Urban, R. Haese, M. Haese, 2005). Examples could be addition, subtraction, multiplication, etc. and point addition. Groups can be defined as following the group axioms (Blythe, Joseph, Urban, R. Haese, M. Haese, 2005):

1. Associativity, meaning for all $a, b, c \in G$ one has $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
2. There exists a unique identity element $e \in G$ such that for all $a \in G$, $e \cdot a = a$ and $a \cdot e = a$.
3. For each element $a \in G$, there exists an element $b \in G$ such that $a \cdot b = e$ and $b \cdot a = e$, where e is the identity element. For all a the element b is unique and is called the inverse of a .

And Abelian groups follow an additional group axiom:

4. Commutativity, meaning for all $a, b \in G$ one has $a \cdot b = b \cdot a$.

Therefore, to prove that $(E, +)$ forms an Abelian group, we must prove that it follows these group axioms, proven by the student in the following theorems.

Theorem 1. $E = \{(x, y) \in \mathbb{F}^2 \mid y^2 = x^3 + Ax + B\} \cup \{\mathcal{O}\}$ has an identity element over point addition $(+)$.

Proof. The identity element is $\mathcal{O}$, as it is a unique point, defined as the point where $P + \mathcal{O} = P$ and $\mathcal{O} + P = P$. $\square$

Theorem 2. Each element in $E = \{(x, y) \in \mathbb{F}^2 \mid y^2 = x^3 + Ax + B\} \cup \{\mathcal{O}\}$ has an inverse in E over point addition $(+)$.

Proof. Let $P \in E = (x_1, y_1)$. P has an inverse $-P \in E = (x_1, -y_1)$. This is true for all points P on E based on symmetry. If $y_1 = 0$, then $P = -P$ and P is self-inverse. For all $P, -P$ we have $P + (-P) = \mathcal{O}$ and $(-P) + P = \mathcal{O}$, where $\mathcal{O}$ is the identity element. $\square$

Theorem 3. $E = \{(x, y) \in \mathbb{F}^2 \mid y^2 = x^3 + Ax + B\} \cup \{\mathcal{O}\}$ is associative over point addition $(+)$.

Proof. From the definition of $\mathcal{O}$ we know $P + Q + R = \mathcal{O}$ from equation (2), it follows from theorems 1 and 2 that:

$$(P + Q) + R = (-R) + R = \mathcal{O} = P + (-P) = P + (Q + R)$$

Hence the group law of associativity is followed. $\square$

Theorem 4. $E = \{(x, y) \in \mathbb{F}^2 \mid y^2 = x^3 + Ax + B\} \cup \{\mathcal{O}\}$ is commutative over point addition $(+)$.

Proof. A line through P and Q is identical to a line through Q and P . Hence, we have $P + Q = Q + P$ $\square$

Hence $(E, +)$ forms an Abelian group.

6. Group Properties

Having proven that $(E, +)$ is an Abelian group, certain group properties can now be established. These properties are the reason elliptic curve groups are suitable for cryptography.

The order of a group $(G, \cdot)$ is the number of elements in it and can be denoted as $|G|$ (Blythe, Joseph, Urban, R. Haese, M. Haese, 2005). For ECDH we must calculate the order of elliptic curves over finite fields, which can be done by using a point counting algorithm, for example Schoof's algorithm (Schoof, 1995). This lays outside of the scope of this essay.

A group $(H, \cdot)$ is a subgroup of a group $(G, \cdot)$, provided that $H \subseteq G$ and H forms a group over $\cdot$ (Blythe, Joseph, Urban, R. Haese, M. Haese, 2005). As $G \subseteq G$, $(G, \cdot)$ is a subgroup of itself. The group $(e, \cdot)$, where e is the identity element is also a subgroup of $(G, \cdot)$ as $e \subseteq G$ and e forms a group over $\cdot$. Every group has the two subgroups described above, any subgroups other than these two are called proper subgroups.

A cyclic group is a group $(G, \cdot)$ where there exists $g \in G$ such that for all $x \in G$, $x = m \cdot g$ for some $m \in \mathbb{Z}$. g is said to be the generator of the group (Blythe, Joseph, Urban, R. Haese, M. Haese, 2005).

We can make cyclic subgroups of our elliptic curve group $(E, +)$ of order n by taking a generator point $G \in E$ and adding it to itself over and over. We will be left with a set $S \subseteq G$ and $S = G, 2 \cdot G, 3 \cdot G, \dots, (m-1) \cdot G$, where m is the order of our subgroup. For ECDH we are interested in finding the order of this subgroup quickly, i.e. without checking every integer $m < n$. We can do this using Lagrange's theorem, taken from Roth (2001), the proof for which is complex and is out of scope of this essay.

Theorem 5 (Lagrange). *The order of a subgroup of a finite group $(G, \cdot)$ is a factor of the order of $(G, \cdot)$.*

We calculate the order $n = |E|$ of $(E, +)$, using Schoof's algorithm (Schoof, 1995), which counts the number of points on E . The details of Schoof's algorithm are outside the scope

of this essay. Using Lagrange's theorem, we only have to check m for each of the factors of n .

We also define the cofactor $h = \frac{n}{m}$ (Housni, 2018), and this will always be an integer as m is a factor of n . Both the cofactor and order of groups formed from points will be important in sections 9 and 10.

As the important properties of elliptic curves have been discussed, now the Elliptic-Curve-Diffie-Hellman protocol will be described.

7. Domain Parameters of ECDH

Domain parameters are variables which are public and preset. Despite there being many possibilities for each parameter, standard sets of values tend to be used as they offer guaranteed security and do not require any verification as they are already verified as strong.

The domain parameters for ECDH are the values (p, a, b, G, n, h) (Jancar, 2018). p is the prime number defining the finite field $\mathbb{F}_p$. a, b are integers defining the elliptic curve E as $y^2 = x^3 + ax + b$, which is over the finite field $\mathbb{F}_p$. G is a generator point on E . n is the order of the cyclic sub-group formed from G . h is the cofactor between the group formed by E and the cyclic subgroup formed from G , and ideally $h = 1$.

How to decide on these domain parameters and why $h = 1$ is ideal will be covered in section 10.

8. Diffie-Hellman Key Exchange

Once domain parameters are established, a Diffie-Hellman key exchange can be performed according to Diffie and Hellman (1976). For this we will need two parties, Alice and Bob, who will use ECDH to communicate privately from third-parties.

Each party has a private key d_A for Alice and d_B for Bob. This is never shared between parties, thus cannot be intercepted by third-parties. Both d_A and d_B are random integers between 1 and $n - 1$, where n is the order of G from the domain parameters.

Each party also has a public key Q_A for Alice and Q_B for Bob. This will be shared between parties, thus can be intercepted by third-parties. Q_A is a point calculated by Alice using her private key and the generator point from the domain parameters as $Q_A = d_A \cdot G$ (Note: here and henceforth $\cdot$ represents scalar multiplication of point addition $+$). Bob similarly calculates $Q_B = d_B \cdot G$.

Now, Alice and Bob swap public keys and use each others public key to calculate a shared secret of x_k . A shared secret is a piece of information known only by Alice and Bob and unknowable to an intercepting third-party. Alice calculates the point $(x_k, y_k) = d_A \cdot Q_B$ and can discard y_k as we are only interested in having a single shared secret. Bob performs the same operation to calculate $(x_k, y_k) = d_B \cdot Q_A$. In both cases, (x_k, y_k) is equal because $d_A \cdot Q_B = d_A \cdot d_B \cdot G = d_B \cdot d_A \cdot G = d_B \cdot Q_A$.

The shared secret is secure provided that both Alice and Bob's private keys cannot be found from their public keys. This is called the elliptic curve discrete logarithm problem (ECDLP), which will be covered in section 9.

Once a shared secret is established between two parties, there are various methods of encoding plain text (Martinez, Encinas, Avila, 2010); however, this is unrelated to the research question, therefore is outside of the scope of this essay. We can assume that once a shared secret is established, communication between two parties is secure.

9. Elliptic Curve Discrete Logarithm Problem

The elliptic curve discrete logarithm problem (ECDLP) is a version of the discrete logarithm problem for elliptic curves. According to Koblitz and Menezes (2004), it becomes: given two points on an elliptic curve G and $P = nG$ over a prime finite field $\mathbb{F}_p$ find n .

It is clear to see the difficulty of this as we must attempt to reverse point addition. The fastest methods for doing this include Pollard's rho algorithm (Teske, 2001) or the Pollig-Hellman algorithm (Teske, 1999). The details of these algorithms are complex and outside of the scope of this essay, however, it is important to note that they run in time proportional to the square root of the order of the group generated by G . This means that they run in half-exponential time with regards to the number of digits of the order, which is not suitably fast to solve the ECDLP where the order is sufficiently high. This is why a cofactor of 1 is ideal, as the order of the subgroup generated from G will be the highest possible, as it will be the same as the order of $G(E, +)$.

Shor's algorithm is a quantum computing algorithm, which can solve discrete logarithm problems in polynomial time with regards to the number of digits in the order of the subgroup generated from G (Shor, 1997). This is fast enough to make ECDH nonviable. Again, the details of Shor's algorithm are complex and outside of the scope of the essay. The only reason that ECDH is still secure today is simply because the computing power of modern quantum computers is not high enough to perform Shor's algorithm (Webber, Elfvig, Weidt, Hensinger, 2022).

10. Selecting Domain Parameters

When selecting domain parameters the weaknesses of ECDH need to be considered. An easy abstraction from doing so would be to choose a standard curve and domain parameters from, for example, the NIST curves (Chen, Moody, Randall, Regenscheid, Robinson, 2023). This may be easier to do, however, it is important to note that some intelligence agencies provide purposely weak curves, where only they are aware of the weakness. This is called a backdoor, and allows these intelligence agencies to decrypt all messages sent using their provided curve, compromising individual security in the name of national security.

In this section I will instead describe what common weaknesses need to be considered and a general approach to deciding on strong domain parameters.

First, when selecting domain parameters, we must ensure that the generator point G has a high order. This follows from the algorithms for solving the ECDLP, as they are half-exponential with regards to the order of G , so a higher order will take longer for the ECDLP to be solved, essentially making it unsolvable in any reasonable amount of time. For G to have a high order we need p to be a large prime to allow for there to be a high number of points on E over $\mathbb{F}_p$. We also need for the order of G to be close, ideally equal, to the order of E , so that all of the large number of points on E increase the order of G . This means having a low cofactor, ideally $h = 1$.

An example of how high the order n of G and the prime p are we can look at the NIST curve P-256 (Chen, Moody, Randall, Regenscheid, Robinson, 2023), which has:

$$\begin{aligned} p &= 115792089210356248762697446949407573530 \\ &\quad 086143415290314195533631308867097853951 \\ n &= 115792089210356248762697446949407573529 \\ &\quad 996955224135760342422259061068512044369 \end{aligned}$$

Next, we must ensure that n is minimally composite, ideally prime, because the Pollig-Hellman algorithm can solve the ECDLP faster for composite orders (Teske, 1999).

We also need our curve E to not be an anomalous curve. An anomalous curve is a curve E over $\mathbb{F}_p$ where:

$$|E| = p$$

This is due to Smart's algorithm, which can solve ECDLP on anomalous curves in polynomial-time using what is called an additive transfer (Galbraith and Smart, 2001). The details of this are complex and outside of the scope of this essay.

There are more weaknesses to be considered when choosing domain parameters, for example Weil and Tate pairing attacks (Koblitz and Menezes, 2004). However, they are less common and more complex, and hence are less important in answering my research question and outside of the scope of the essay. There are also weaknesses regarding how ECDH is implemented in a system, some outlined by Jancak (2018), which should be considered when deciding on domain parameters. These are again outside of the scope of this essay. It should also be noted that new weaknesses are constantly being found and studied, hence covering them all is not possible.

Once we know what parameters to avoid, we must be able to generate desirable parameters. One method of doing so is called the exhaustive approach or the random approach (Jancak, 2018). This consists of randomly choosing an elliptic curve and a prime finite field, domain parameters a, b and p , from a desired range of numbers. From here, the number of points on the curve can be counted using Schoof's algorithm to find the order of the curve group. Then the point G is chosen and its order is calculated similarly. and the cofactor h can easily be calculated after. During this process and after the curve is tested for any known weaknesses and if found the parameters are consequently thrown out and the process repeated.

Despite this method being inefficient, it is used as the domain parameters produced are verifiably random, convincing the users that no backdoor was purposefully hidden in them. The inefficiency is not a large problem, as new domain parameters do not need to be constantly generated when using ECDH.

Despite how meticulously domain parameters for ECDH are chosen, there is one fatal,

inevitable weakness common to them all. In the previous section it was noted that Shor's algorithm (Shor, 1997), a quantum algorithm, has been proven to be able to solve the ECDLP in polynomial-time. This simply cannot be avoided, and as quantum computers become more powerful, modern cryptography will require drastic changes, ECDH and RSA, and any other current cryptography system reliant on the discrete logarithm problem will no longer provide online security. Nonetheless, elliptic curves still remain a promising area of study for cryptography. A new, quantum-proof cryptography system using elliptic curve isomorphism groups is currently being developed (Feo, Jao, Plut, 2014).

11. Conclusion

The properties of elliptic curves can be used in the ECDH protocol to achieve online security. This security relies solely on the elliptic curve discrete logarithm problem, which can be solved in polynomial-time using Shor's algorithm. It is only a matter of time as to when the number of q-bits in quantum computers grows to a sufficient number to allow Shor's algorithm to be run, thus it is only a matter of time until ECDH can no longer provide online security. This will however be quite some time as modern quantum computers have so little processing power (Sparkes, 2021). The quantum-computing revolution does not uniquely target ECDH; RSA and many of the most popular current-day cryptography systems rely on the discrete logarithm problem and its variants.

In current cryptography it is seen as certain that the security of these cryptography systems will not last for much longer. For this reason, new, quantum-proof cryptography systems are being developed. One such system is based on supersingular elliptic curve isogeny graphs, showing again how useful elliptic curves are. But, until quantum computers have enough q-bits to run Shor's algorithm, ECDH remains one of the best approaches to achieving online security between two parties.

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